AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX

ANNUAL EXAMINATIONS (THEORY) 2024

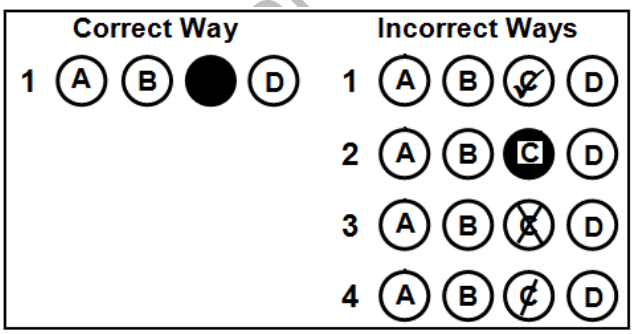
Biology Paper I

Time: 1 hour 10 minutes Marks: 40

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 40 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. Consider the given image.



The researcher in the given image can be identified as a

A. histologist.
B. taxonomist.
C. palaeontologist.
D. biotechnologist.

2. In contrast to multicellular organisms, unicellular organisms

A. show division of labour at organ system level.
B. are entirely exposed to the environment.
C. have complex body organisation.
D. always reproduce sexually.

3. Consider the following observations.

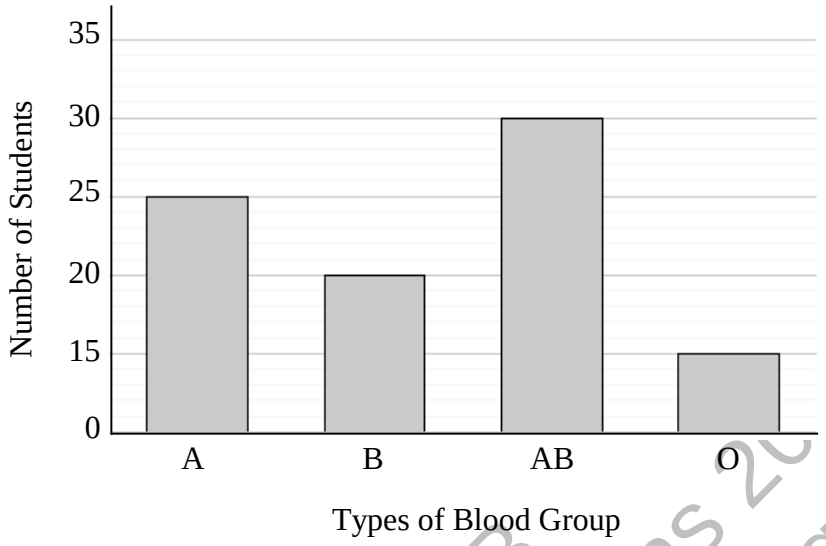
I. The temperature of a room increases by 5 °C.
II. Flowers in an orchard have round shaped petals.
III. The growth of fungus on a piece of bread indicates that it is stale.
IV. The rate of reaction increases twice by increasing the enzyme concentration.

The CORRECT pair of qualitative observations is

A. I and II.
B. I and IV.
C. II and III.
D. III and IV.

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4. The given graph shows the number of students having different blood groups in a class.



If three students with blood group A and four students with blood group B leave the class while two students with blood group O join the class, then which blood group students are present least in number?

A. Blood Group O
B. Blood Group A
C. Blood Group B
D. Blood Group AB

5. The given table shows structural characteristics of an organism.

Property	Present or Absent
Cell wall	Present
Nuclear membrane	Present
Ribosomes	Present
Multi-cellularity	Absent

Based on the given characteristics, the organism could be placed in kingdom

A. Protista.
B. Monera.
C. Plantae.
D. Fungi.

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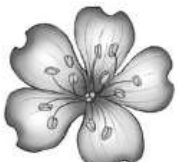
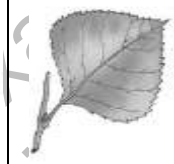
6. Consider the given table.

Unit of Classification	Example
Kingdom ---- Animalia	housefly, butterfly, whale, frog, snake, bush fly, fruit fly, mayfly
Phylum ---- Arthropoda	housefly, butterfly, fruit fly, bush fly, mayfly
Genus ---- Musca	housefly, bush fly
Species ---- domestica	housefly

According to the given classification, the animal which shares the MOST common characteristics with housefly is a

A. mayfly.
B. fruit fly.
C. bush fly.
D. butterfly.

7. The set of structures, which represent monocot plants, is

					
I	II	III	IV	V	VI

A. I, III and V.
B. I, IV and V.
C. II, IV and VI.
D. II, III, and VI.

8. In 2011, Western black rhinoceros were declared as critically endangered animals due to their over-hunting.

Which of the following conservation measures is the MOST effective in preventing the rhinoceros from extinction?

A. Vaccinating them regularly
B. Introducing different species in their habitat
C. Attaching metal tags to identify them as endangered animals
D. Establishing sanctuaries where human interference would be limited

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9. Which of the following CORRECTLY identifies the organelle shown in the diagram?



A. Ribosome
B. Chloroplast
C. Mitochondria
D. Endoplasmic reticulum

10. In the cell, Golgi apparatus lies near the endoplasmic reticulum because Golgi apparatus

A. provides mechanical strength to the endoplasmic reticulum.
B. receives and utilises ATP molecules from the endoplasmic reticulum.
C. modifies and packages proteins produced by the endoplasmic reticulum.
D. controls the movement of substances in and out of the endoplasmic reticulum.

11. Which of the following options exemplifies the process of exocytosis in a cell?

A. Phagocytosis of bacteria by white blood cells
B. Secretion of digestive enzymes by pancreatic cells
C. Movement of glucose against the gradient with the help of membrane-bound proteins
D. Movement of sodium and potassium ions from a neuron through sodium-potassium pump

12. The specimens that can be observed under electron microscope are

I. a sample of epithelial layer
II. a culture of reproducing bacteria
III. a moving amoeba in a drop of water
IV. a piece of tissue taken from mammary gland

A. I and II.
B. I and IV.
C. II and III.
D. III and IV.

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13. The given table shows function of structures, P, Q and R in living things.

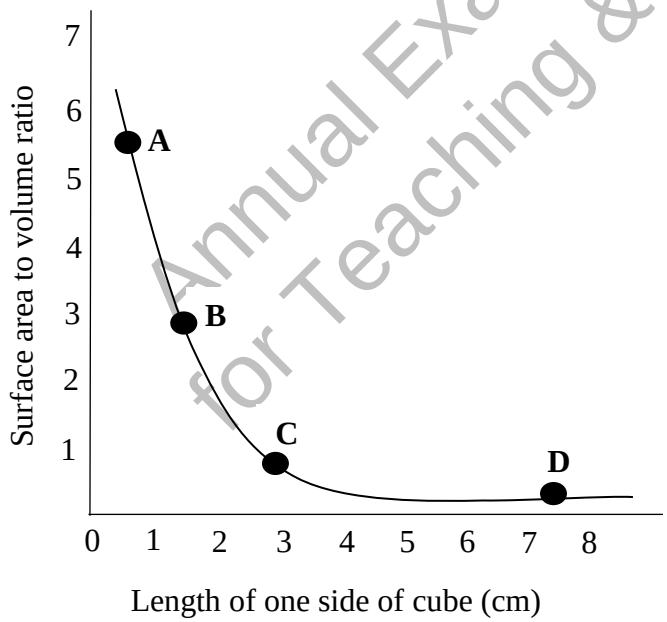
Structure	Function
P	Transport water molecules
Q	Transport oxygen molecules
R	Absorb surrounding water through osmosis

The option which CORRECTLY identifies the structures **P**, **Q** and **R** is

	P	Q	R
A	xylem vessels	red blood cells	root hair cells
B	red blood cells	root hair cells	xylem vessels
C	root hair cells	xylem vessels	red blood cells
D	root hair cells	red blood cells	xylem vessels

14. Jamal calculated the surface area to volume ratio of four different sizes of potato cubes (A, B, C and D) and he plotted the given graph to analyse the results.

With reference to the graph, the potato cube that has more efficient rate of diffusion is



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15. Consider the given chemical reaction of water breakdown.

$H_2O \rightarrow 2H^+ + 2e^- + 1/2O_2$

The electrons resulted in the given reaction are gained by

A. CO_2 in the carbon fixation.
B. O_2 during cellular respiration.
C. chlorophyll in the photosystem.
D. ADP in electron transport chain.

16. Suppose ATP synthesis is stopped in an illuminated chloroplast. This may happen due to

A. damaged thylakoid membrane.
B. denaturation of enzymes present in stroma.
C. excessive absorption of light by chloroplast.
D. excessive transfer of protons from stroma to thylakoid space.

17. When a molecule of adenosine diphosphate reacts with water, it gives

A. adenosine triphosphate and hydrogen gas.
B. adenosine monophosphate and hydrogen gas.
C. adenosine triphosphate and inorganic phosphate.
D. adenosine monophosphate and inorganic phosphate.

18. Consider the given word equations.

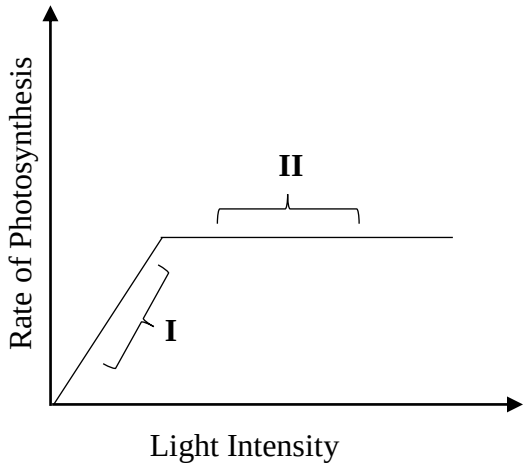
I. glucose → lactic acid + energy
II. glucose → ethanol + carbon dioxide + energy
III. glucose + oxygen → carbon dioxide + water + energy
IV. carbon dioxide + water + energy → glucose + oxygen

The option that shows the equation for anaerobic respiration in humans and in yeast is

	Anaerobic Respiration in Human	**Anaerobic Respiration in Yeast**
A	I	II
B	I	III
C	II	IV
D	III	IV

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19. The given graph shows one of the factors affecting rate of photosynthesis.



The option that CORRECTLY identifies the limiting factor(s) in the labelled parts **I** and **II** is

	I	**II**
A	light intensity	carbon dioxide and temperature
B	carbon dioxide and temperature	light intensity
C	light intensity and carbon dioxide	water
D	water	light intensity and carbon dioxide

20. Calvin cycle occurs in the stroma of chloroplast because the stroma contains

A. required enzymes.
B. DNA and ribosomes.
C. photosynthetic pigments.
D. ATP synthase complexes.

21. In the human respiratory system, the function of epiglottis is to

A. moist and warm the incoming air.
B. keep the air ways open all the time.
C. filter dust particles and kill the bacteria.
D. prevent the food particles from entering the trachea.

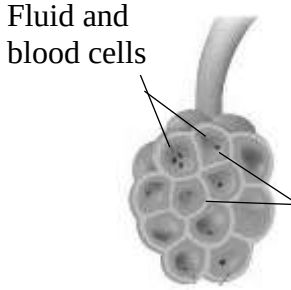
22. In the human respiratory system, the hypersecretion (increased production) of mucus causes development of respiratory diseases such as bronchitis.

All of the following factors cause over-production of mucus EXCEPT

A. viral infections.
B. bacterial infections.
C. malfunctioning of cilia.
D. hyperactivity of goblet cells.

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23. The given images show the comparison of healthy and diseased alveolar sacs (**I** and **II**) in humans.

		
Alveolar Sac I	**Healthy Alveolar Sac**	**Alveolar Sac II**

The alveolar sacs **I** and **II** represent

	Alveolar Sac I	**Alveolar Sac II**
A	emphysema	pneumonia
B	pneumonia	emphysema
C	asthma	bronchitis
D	bronchitis	asthma

24. The given steps occur during breathing mechanism in the human respiratory system.

I. Size of chest cavity decreases
II. Pressure in the lungs decreases
III. The muscles of diaphragm relax
IV. Ribs move upwards and outwards

The steps which take place during exhalation are

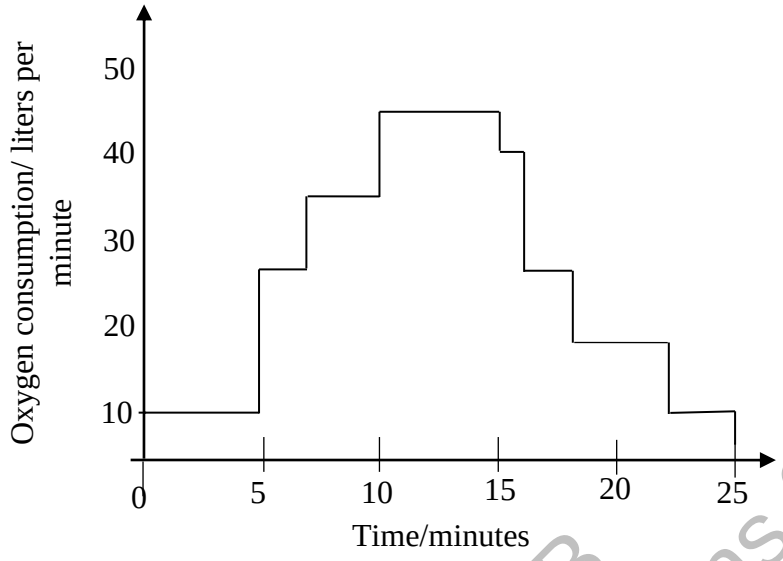
A. I and II.
B. I and III.
C. II and IV.
D. III and IV.

25. The chemical reaction that requires extra-cellular enzymes is

A. breakdown of fats.
B. oxidation of glucose.
C. synthesis of proteins.
D. reduction of carbon dioxide.

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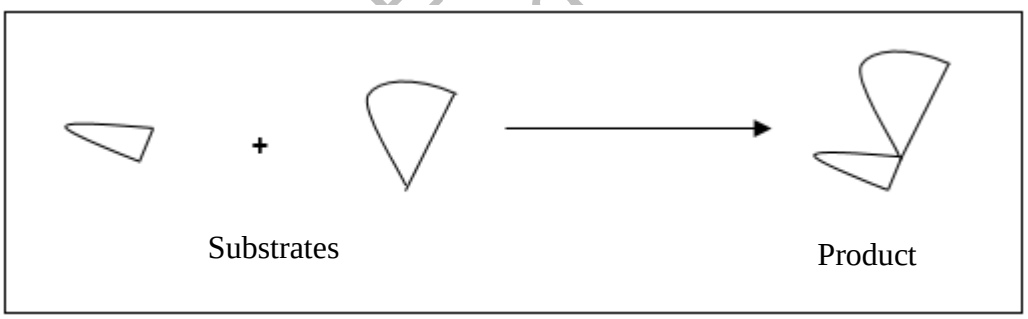
26. The given graph shows the oxygen consumption during and after exercise.



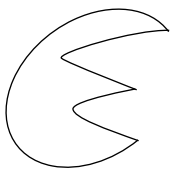
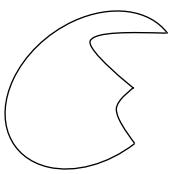
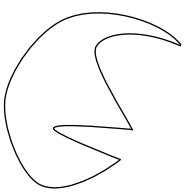
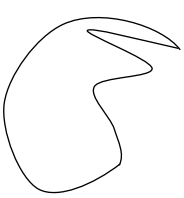
With reference to the given graph, the period of strenuous exercise would be from

A. 5 to 10 minutes.
B. 10 to 15 minutes.
C. 15 to 20 minutes.
D. 20 to 25 minutes.

27. The given diagram represents an enzyme-catalysed reaction.

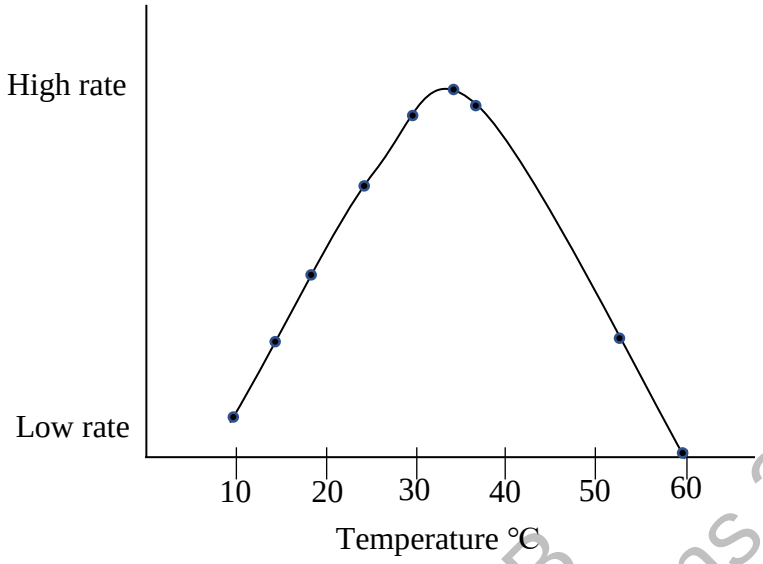


With reference to lock and key model, the enzyme that can catalyse this reaction is

			
A	B	C	D

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28. The given graph shows the effect of temperature on the enzyme activity.



The temperature range at which the substrate molecules will no longer fit into enzyme is

A. 10 – 20 °C.
B. 20 – 30 °C.
C. 30 – 40 °C.
D. 50 – 60 °C.

29. Which of the following indicates the movement of larynx and state of trachea during the swallowing of food?

	Movement of Larynx	**Passage to Trachea**
A	Up	Open
B	Up	Closed
C	Down	Open
D	Down	Closed

30. In plants, the molecule that will be DIRECTLY affected by the deficiency of magnesium ions in soil is

A. starch.
B. cellulose.
C. chlorophyll.
D. carbon dioxide.

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31. What will happen if the human stomach is unable to secrete hydrochloric acid?

A. Lipid digestion will inhibit
B. Amylase enzyme will denature
C. Stomach muscles will destruct
D. Pepsinogen will remain inactive

32. Following are diseases that are related to dietary habits.

- Heart diseases
- Constipation
- Inflammation of colon
- Irritated bowl syndrome

Which of the following diets is MOST likely to develop the given diseases?

A. A diet with low fiber
B. A diet with low sodium level
C. A diet with moderate sugar content
D. A diet with moderate unsaturated fats

33. Zaheer's gut is unable to absorb mineral salts and water from the undigested food because of an abnormality in his

A. stomach.
B. oesophagus.
C. large intestine.
D. small intestine.

34. Which of the following CORRECTLY identifies the site of action of trypsin and pepsin in the alimentary canal of human?

	Trypsin	**Pepsin**
A	Ileum	Colon
B	Stomach	Duodenum
C	Colon	Ileum
D	Duodenum	Stomach

35. The cells that transport organic nutrients from leaves to the other parts of a plant are

A. vessels and tracheids.
B. sieve tubes and tracheids.
C. companion cells and vessels.
D. sieve tubes and companion cells.

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36. During the circulation of blood through atria and ventricles of human heart, atrioventricular valves close when blood pressure is

A. higher in atria than aorta.
B. higher in ventricles than atria.
C. lower in aorta than pulmonary arteries.
D. lower in ventricles than pulmonary arteries.

37. In plants, excessive transpiration causes

A. loss of turgidity in leaves cells.
B. increase in the cellular respiration.
C. increase in the rate of photosynthesis.
D. loss of water from phloem to xylem tissues.

38. The blood vessel which carries deoxygenated blood from lower body parts to heart is

A. pulmonary vein.
B. inferior vena cava.
C. hepatic portal vein.
D. superior vena cava.

39. The given table shows blood reports of four individuals (**A**, **B**, **C** and **D**).

With reference to changes in the count of blood cells, the blood report of the individual that indicates the presence of a bacterial infection is

A

Test	Result	Reference Range
WBCs	14,000/mm^3	4,000-11,000
Neutrophils	88	50-70%
Lymphocytes	13	25-40%

B

Test	Result	Reference Range
WBCs	10,000/mm^3	4,000-11,000
Neutrophils	61	50-70%
Lymphocytes	32	25-40%

C

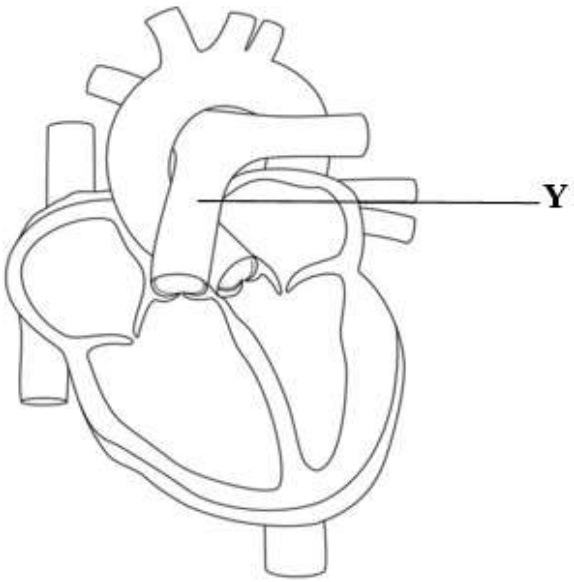
Test	Result	Reference Range
WBCs	11,000/mm^3	4,000-11,000
Neutrophils	40	50-70%
Lymphocytes	20	25-40%

D

Test	Result	Reference Range
WBCs	8,900/mm^3	4,000-11,000
Neutrophils	50	50-70%
Lymphocytes	18	25-40%

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40. The given diagram shows longitudinal section of the human heart.



Which of the following options is CORRECT about the concentration of carbon dioxide and blood pressure in the vessel labelled as **Y**?

	Concentration of Carbon dioxide	**Blood Pressure**
A	High	High
B	Low	High
C	High	Low
D	Low	Low

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